

ECE595: Power Converters for AC Systems

Lecture 25 *(Single-phase B2B Config., Applic. and More)*

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Academic Calendar

Midterm 1: (03/10/2026) Lecture 17

Midterm 2: (04/23/2026) Lecture 28

Project: (04/30/2026)

HW10

Problem 1

Considering the parameters of the DC machine as presented below

$R_a=0.06$;

$L_a=0.0018$;

$k_c=0.1$;

$B_m=0.001$;

$J_m=0.03$;

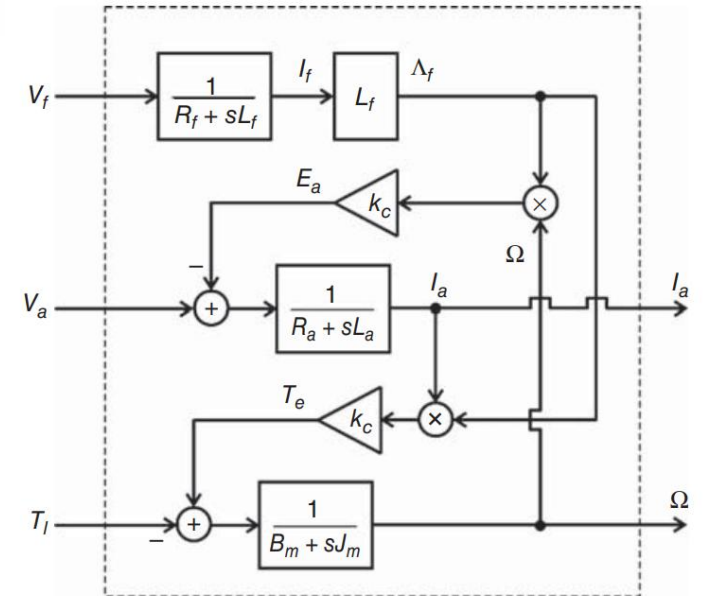
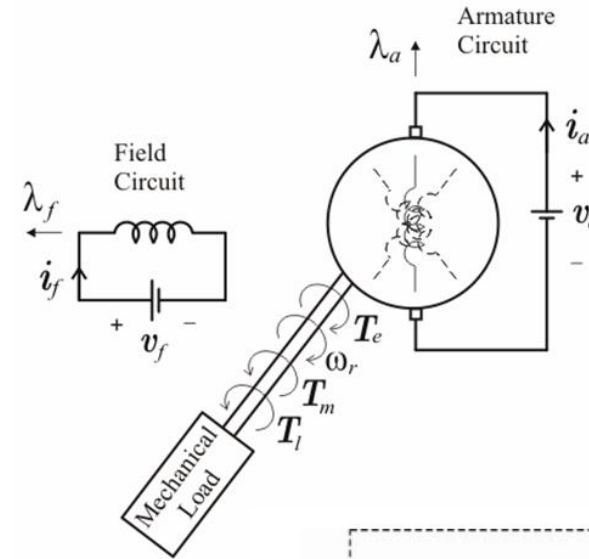
$V_a=1$;

$T_l=0.01$;

Field flux = 1;

Simulate this machine in the time-domain (MATLAB script) and frequency-domain (SIMULINK) and plot both armature current and mechanical speed for each simulation. Compare the results. Simulate enough time to see both waveforms (i_a and ω_m) in steady state conditions.

$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$



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Problem 2

Make the necessary modifications on both codes (t-domain and s-domain) to add a Proportional controller to control armature current of 10 A with gains of $K_p=10$, 100, and 500. Plot the armature current, its reference and mechanical speed.

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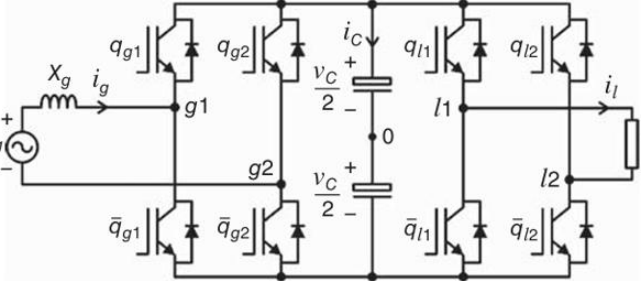
Problem 3

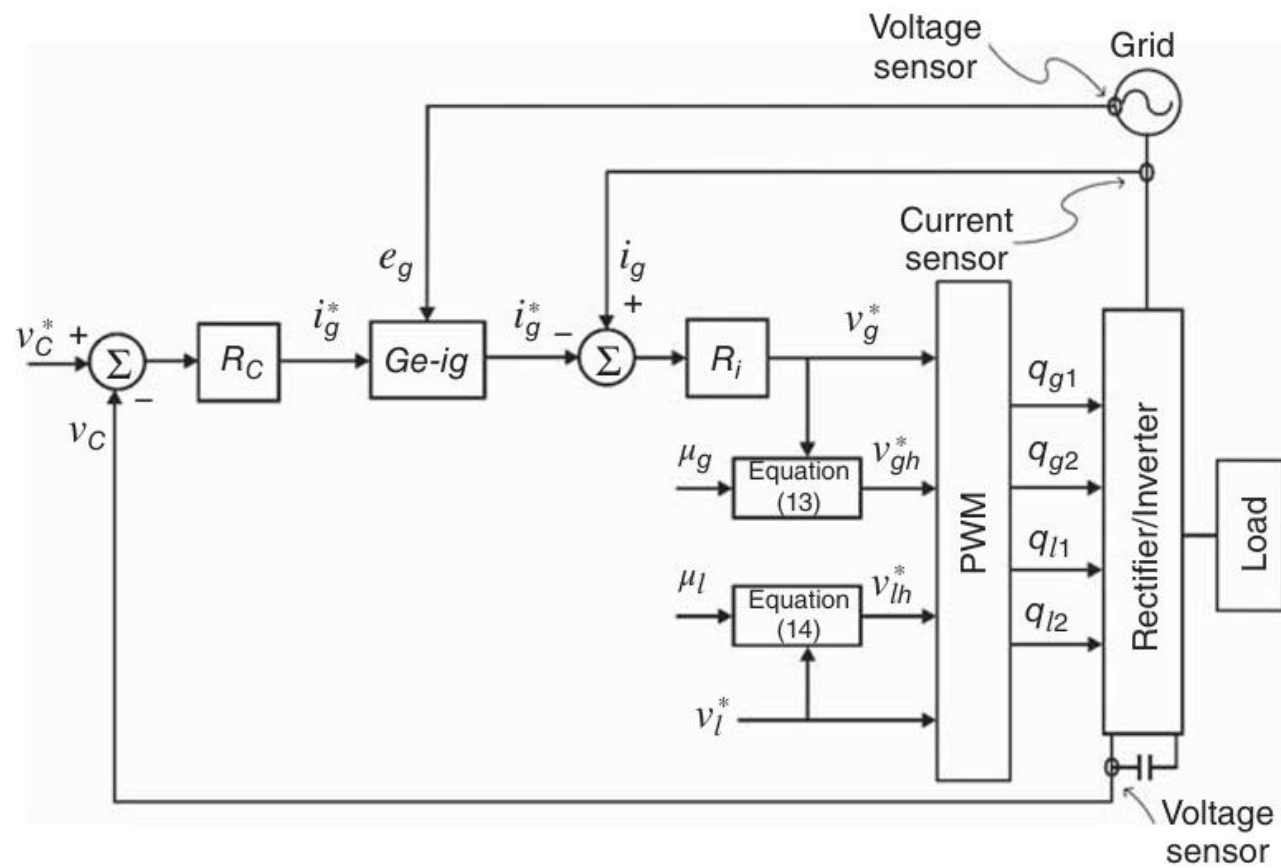
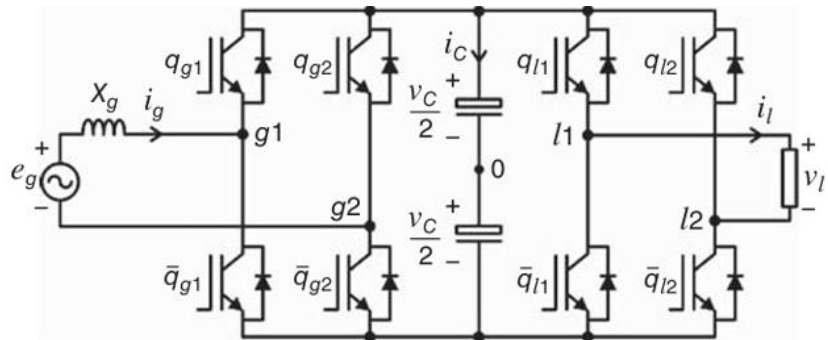
Add an H-bridge power converter for the system presented in Problem 2 with V_{dc} (dc-link voltage) equal to 2 (normalized to satisfy the conditions of the motor), assume a switching frequency of 10 kHz.

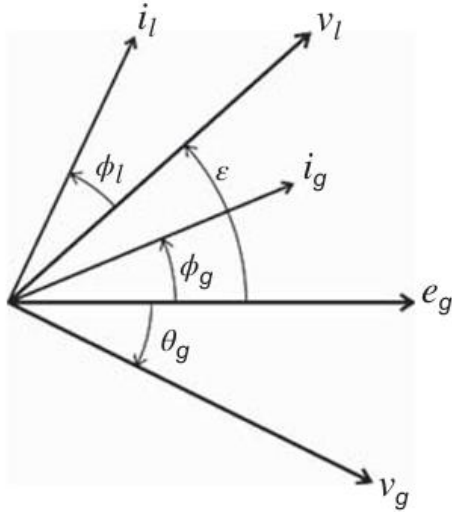
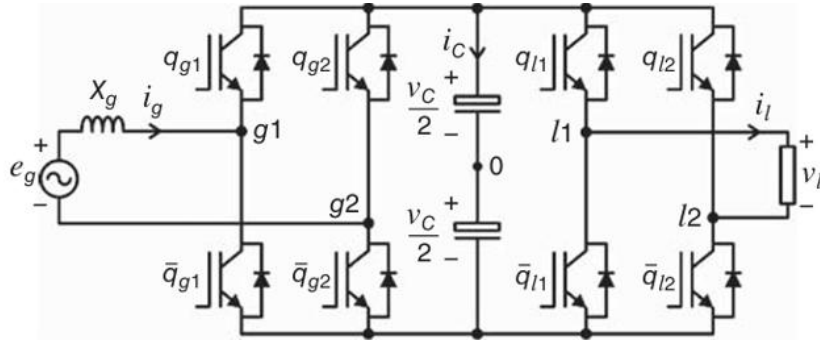
HW10

Problem 4

Simulate a PI current controller to regulate a reference current of 10 A in the previously presented DC machine. The system consists of an H-bridge converter ($V_{dc} = 2$ V, switching frequency $f_s = 10$ kHz) supplying the DC machine. Determine appropriate PI controller gains to ensure zero steady-state error. Plot both the reference current and the measured current.







$$p_c(t) = p_g(t) - p_l(t)$$

$$p_c(t) = \left[\frac{V_g I_g}{2} \cos(\theta_g + \Phi_g) - \frac{V_l I_l}{2} \cos(\Phi_l) \right] + \left[\frac{V_g I_g}{2} \cos(2\omega_g t - \theta_g + \Phi_g) - \frac{V_l I_l}{2} \cos(2\omega_l t + 2\epsilon + \Phi_l) \right]$$

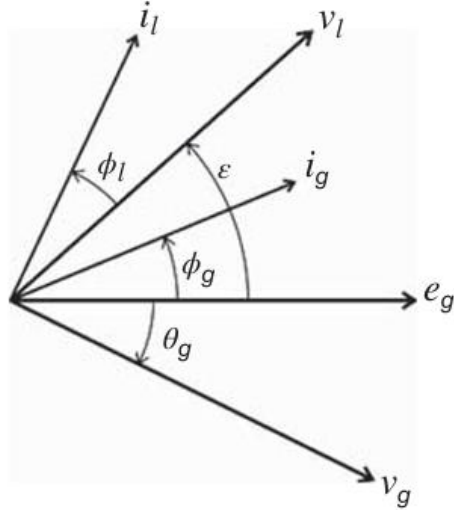
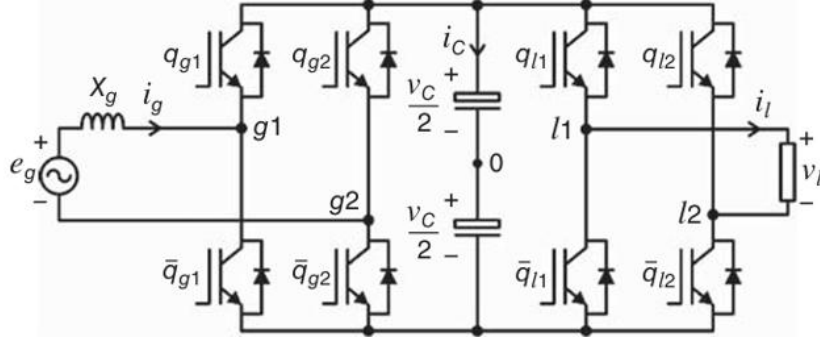
$$V_g I_g \cos(\theta_g + \Phi_g) = V_l I_l \cos(\Phi_l)$$

$$p_c(t) = \frac{V_g I_g}{2} \cos(2\omega_g t - \theta_g + \Phi_g) - \frac{V_l I_l}{2} \cos(2\omega_l t + 2\epsilon + \Phi_l)$$

$$p_c(t) = N_l \frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} \cos(2\omega_g t - \theta_g + \Phi_g) - N_l \cos(2\omega_l t + 2\epsilon + \Phi_l)$$

where $N_l = V_l I_l / 2$.

$$p_c(t) = v_c i_c = C v_c \frac{dv_c}{dt}$$



$$p_c(t) = v_c i_c = C v_c \frac{dv_c}{dt}$$

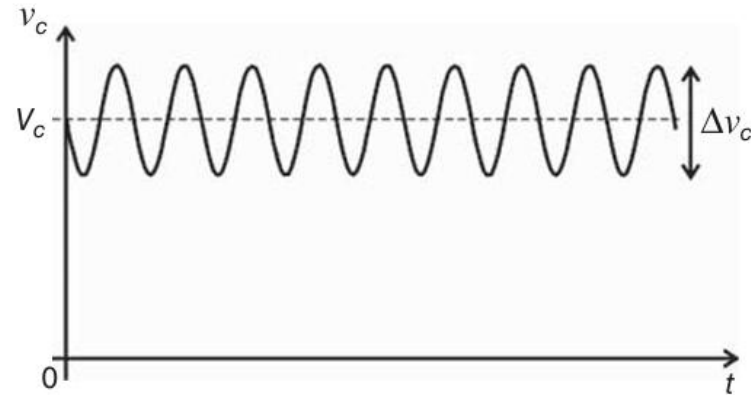
$$v_c \frac{dv_c}{dt} = \frac{N_l}{C} \left[\frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} \cos(2\omega_g t - \theta_g + \Phi_g) - \cos(2\omega_l t + 2\epsilon + \Phi_l) \right]$$

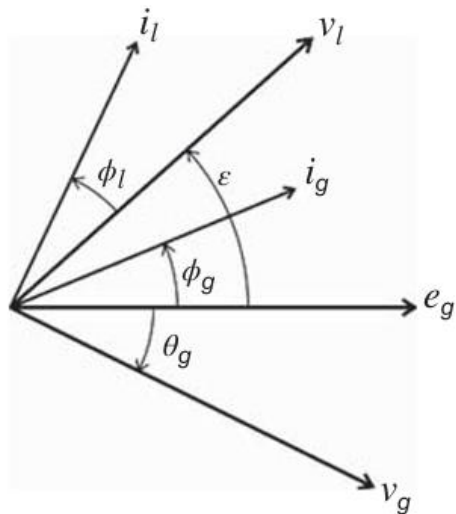
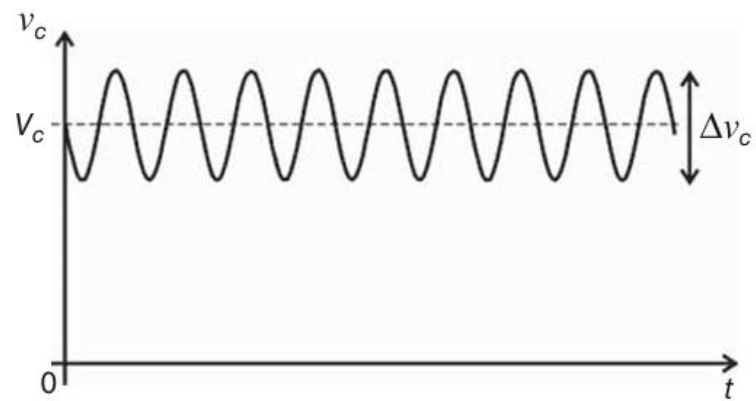
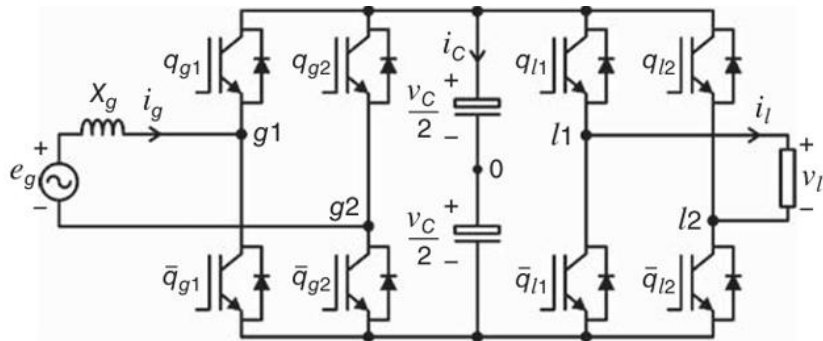
Applying an integral at both sides of equation

$$v_c^2 = N_l \left[\frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} X_{cg} \sin(2\omega_g t - \theta_g + \Phi_g) - X_{cl} \sin(2\omega_l t + 2\epsilon + \Phi_l) \right] + \bar{v}_c$$

$$v_c = \sqrt{\bar{v}_c + \tilde{v}_c}$$

$$\tilde{v}_c = N_l X_{cl} \left[\frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} \frac{f_l}{f_g} \sin(2\omega_g t - \theta_g + \Phi_g) - \sin(2\omega_l t + 2\epsilon + \Phi_l) \right]$$





$$\Delta v_c \simeq \frac{N_l X_c}{V_c} k_\epsilon$$

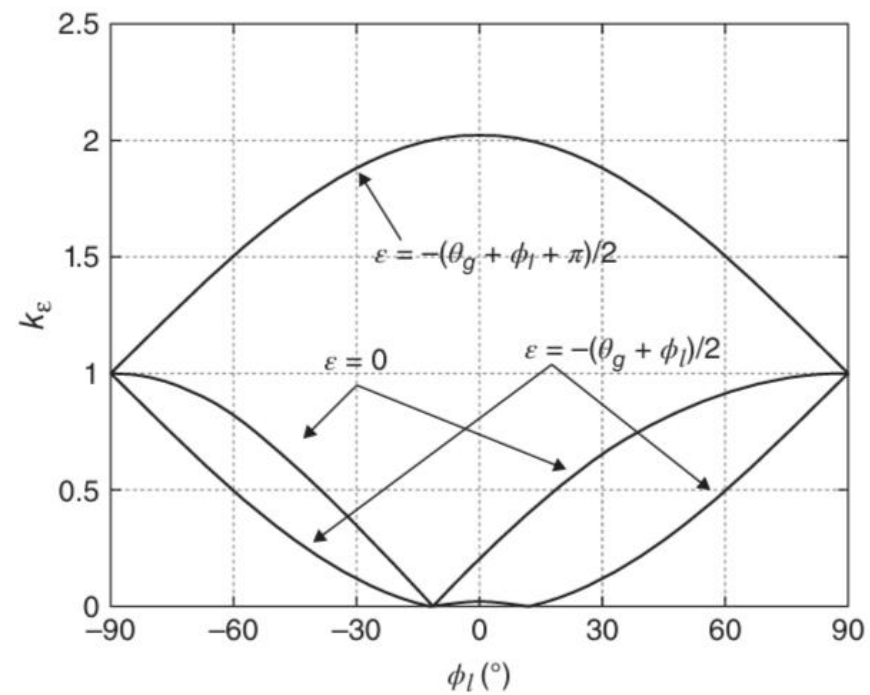
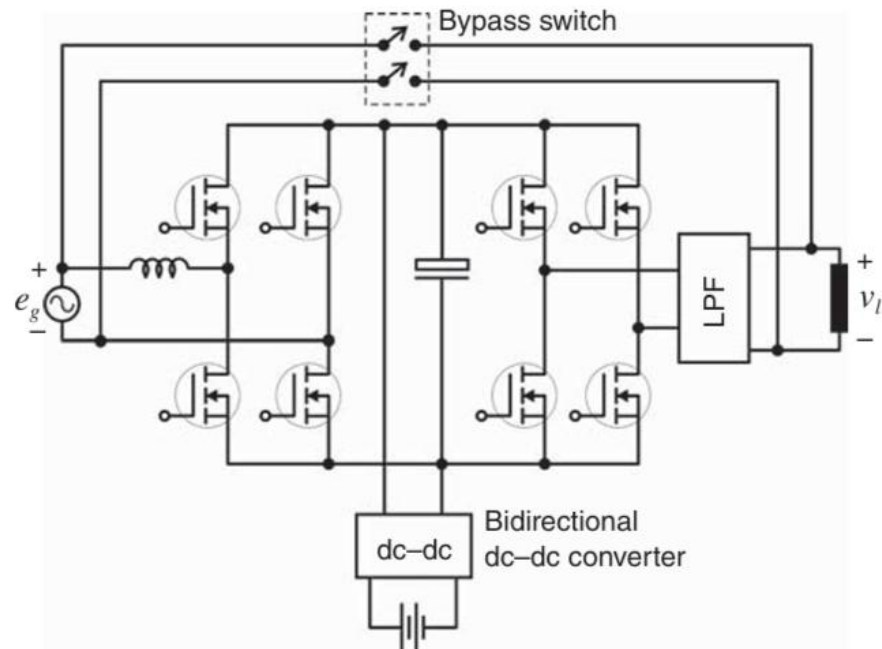
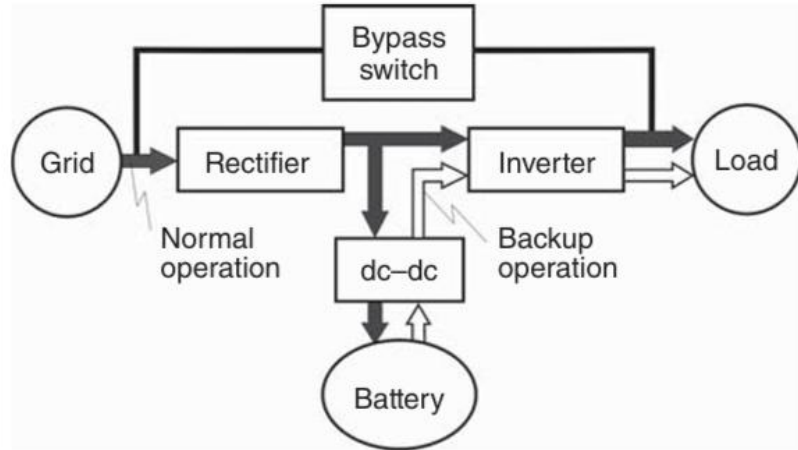
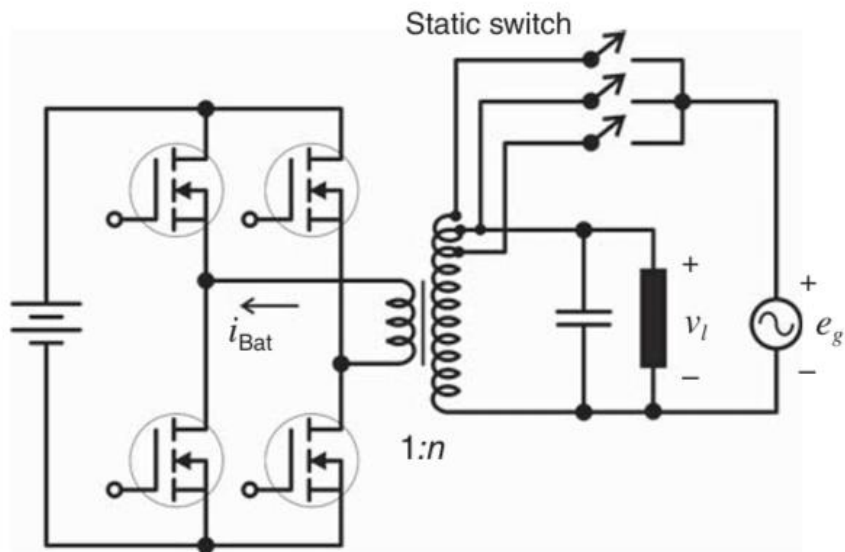
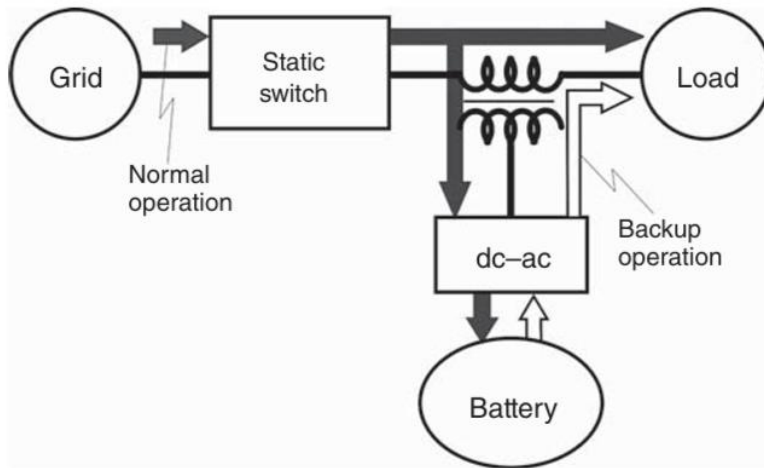


Figure 10.5 k_ϵ as a function of the load power factor ϕ_l .

Application (Uninterrupted Power Supply– UPS)

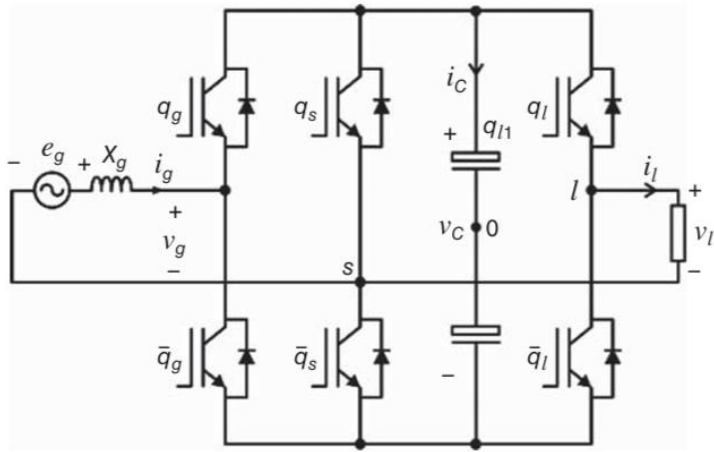
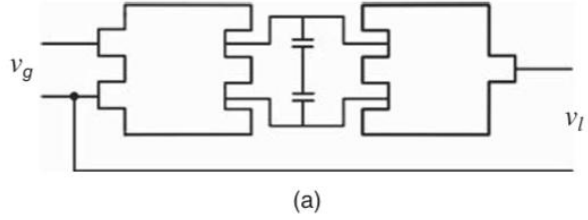


UPS is the line-interactive

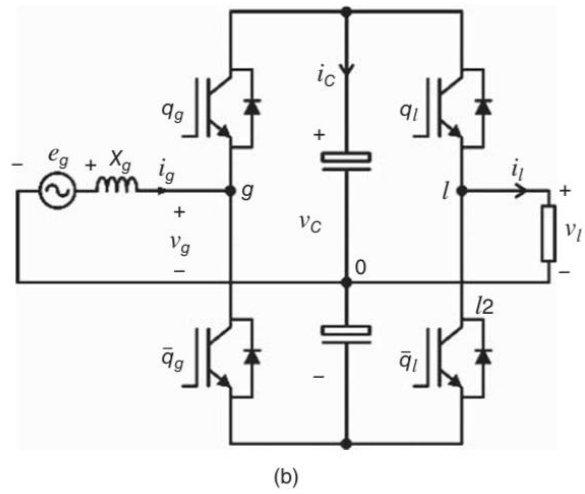
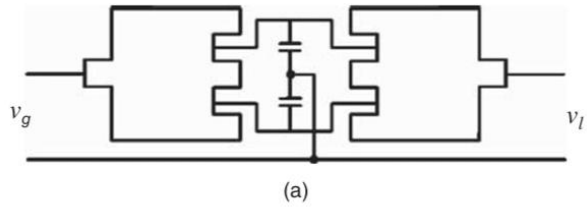


Application – Active Power Filter

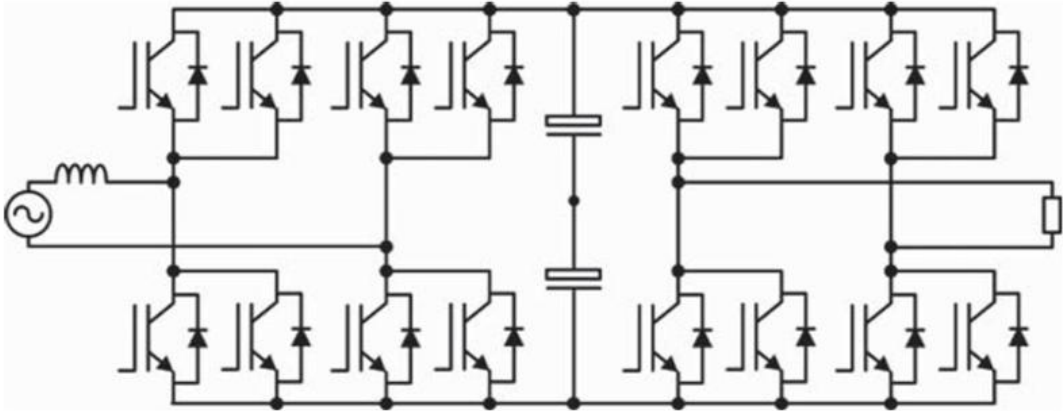
Topology With Component Count Reduction (Shared-leg)



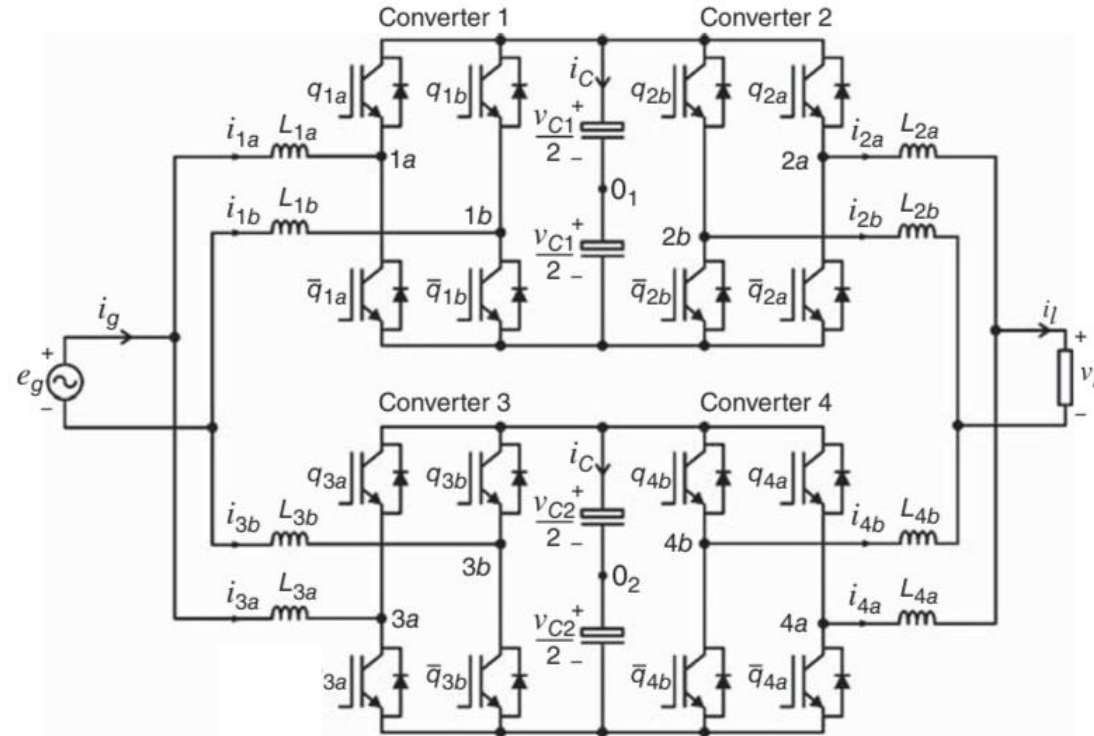
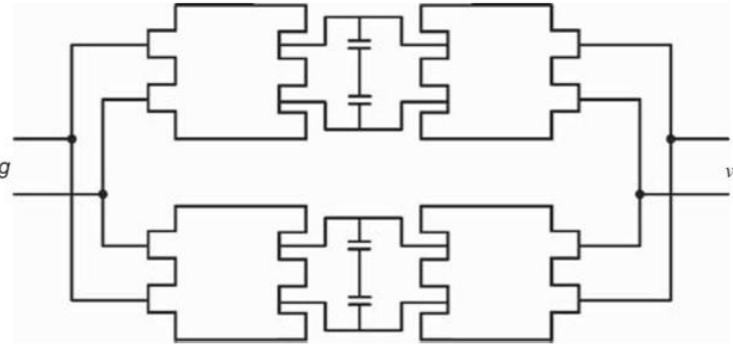
Topology With Component Count Reduction (Half-bridge)

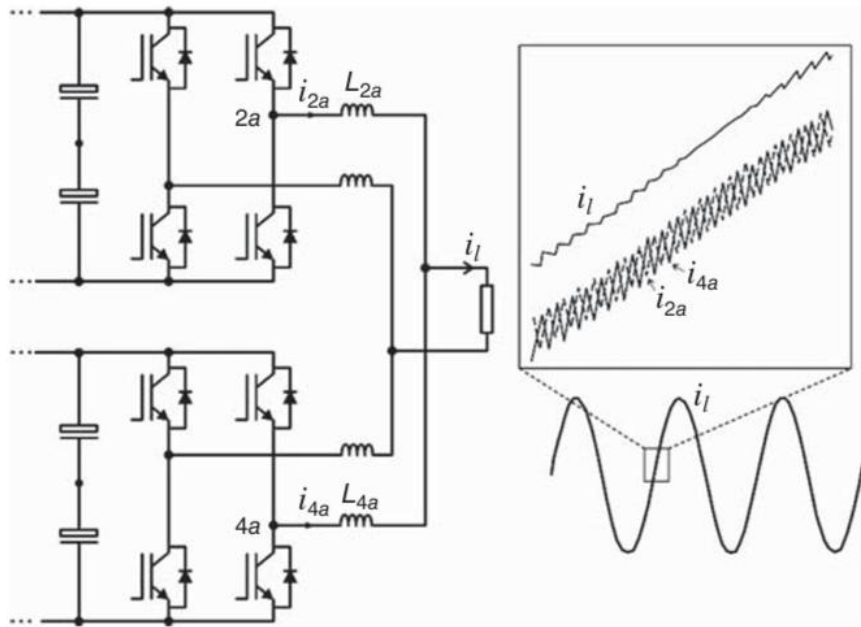


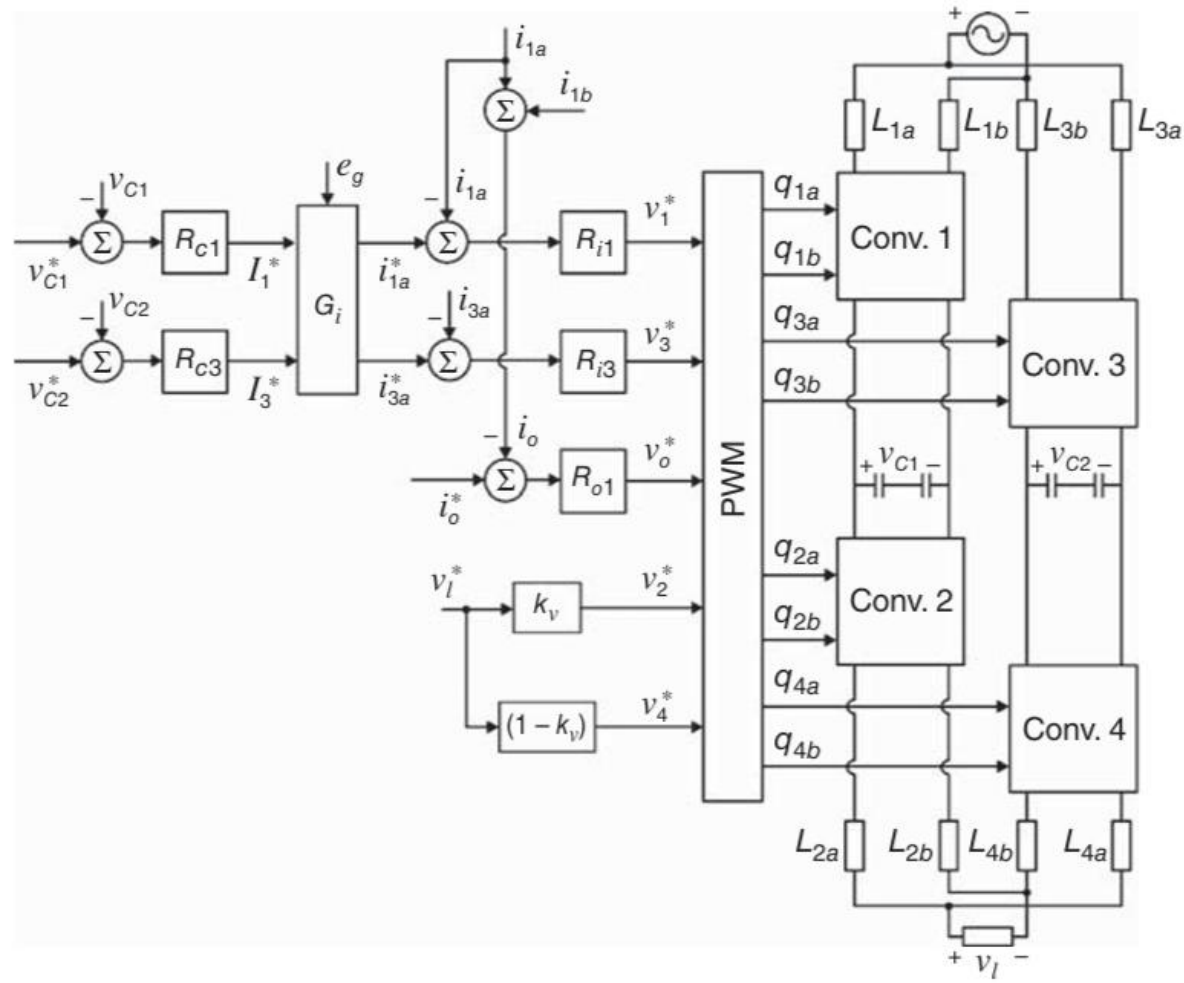
Topology With Increased Number of Switches



Topology With Increased Number of Switches (Converters in Parallel)







Topology With Increased Number of Switches (Converters in Series)

